

Species-Aware Module Review: Fructose PTS Uptake and Entry into Central Carbon Catabolism in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module:** Fructose-specific PEP-PTS import ($\rightarrow$ fructose-1-phosphate) + 1-phosphofructokinase (F1P $\rightarrow$ fructose-1,6-bisphosphate). Boundary ends at fructose-1,6-bisphosphate (FBP). **Seed bucket:** KEGG ppu00051 "Fructose and mannose metabolism" (18 candidate genes)

1. Executive summary

The generic "bacterial fructose PTS uptake $\rightarrow$ F1P $\rightarrow$ FBP" module is **fully satisfiable in *P. putida* KT2440** and maps cleanly and unambiguously to a single, experimentally characterized operon, **fruBKA** (PP_0793 – PP_0794 – PP_0795). All three module activities are encoded with matching EC/GO annotations, supported by direct KT2440 experimental work.

The single most important curation issue is a **metadata gap, not a biological gap**: the phosphorelay gene **fruB** (PP_0793, UniProt Q88PQ5) is **absent from the 18-gene candidate list**. This is an artifact of seeding candidates from KEGG pathway **ppu00051**, whereas KEGG places **fruB** only on the PTS map **ppu02060**. **fruB** should be added and promoted to full review.

Conversely, **15 of the 18 candidate genes are over-propagated** into this module. They are legitimate members of the broad KEGG "Fructose and mannose metabolism" map (alginate biosynthesis, the GDP-mannose branch, fucose catabolism, and gluconeogenesis/triose enzymes) but lie **outside** the fructose-PTS $\rightarrow$ F1P $\rightarrow$ FBP boundary. Only **fruA** and **fruK** from the candidate list are true module members; **fba** and **fbp** are downstream-adjacent context.

2. Target-organism pathway definition

Included process (KT2440-specific): PEP-dependent phosphoryl relay through a fused multiphosphoryl-transfer protein (EI-HPr-EIIA^{Fru} = FruB), transfer to the membrane EIIBC^{Fru} permease (FruA) which imports extracellular fructose and phosphorylates it on import to **fructose-1-phosphate (F1P)**, followed by ATP-dependent phosphorylation of F1P to **fructose-1,6-bisphosphate (FBP)** by 1-phosphofructokinase (FruK). The module ends at FBP.

Key species-specific facts: - KT2440 has **only two PTS branches**: a *complete* fructose-specific PTS (PTS^{Fru} = FruBKA) and the sugar-unrelated nitrogen PTS^{Ntr} (PtsP/PtsO/PtsN). Fructose is the **only** sugar taken up by a PTS in this organism (Chavarría et al. 2012,

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- **Glucose is NOT PTS-transported** in KT2440; it enters via porins/ABC transport and three convergent peripheral routes converging at 6-phosphogluconate, feeding the **Entner-Doudoroff (Edd/Eda)** pathway (del Castillo et al. 2007,

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- Downstream of the module endpoint, KT2440 **lacks 6-phosphofructokinase (Pfk)** — confirmed by KEGG: neither K00850 (ATP-dependent 6-PFK, EC 2.7.1.11) nor K00895 (PPi-dependent PFK) is encoded in `ppu`; the only phosphofructokinase-type gene is K00882 = `fruK` / PP_0794 (1-PFK, EC 2.7.1.56). Net glycolytic (EMP) flux from F6P→FBP therefore does not occur. Fructose-derived FBP is instead routed via FBP aldolase (Fba) to trioses and/or recycled to F6P by fructose-1,6-bisphosphatase (Fbp) to enter the ED pathway — the "EDEMP cycle." This is **outside** the module boundary but explains why `fbp` is physiologically essential for fructose catabolism here. - **Operon context (KEGG-confirmed gene order):** PP_0792 `cra` → PP_0793 `fruB` → PP_0794 `fruK` → PP_0795 `fruA` → PP_0796 (unknown). The Cra/FruR regulator is immediately upstream of the fruBKA operon, consistent with F1P-dependent operon-level regulation (regulation itself is outside the module boundary).

Neighboring pathways/maps to keep separate: ppu02060 (PTS map — where `fruB` lives), ppu00010/ppu00030 (glycolysis/ED/PPP downstream of FBP), ppu00520/ppu00051 mannose & GDP-mannose branch, ppu00543/ppu00040 alginate & fucose. **Alternate names:** PTS^{Fru} / PTS^{Fru} / fructose PTS; the operon is also cited as *fruBKA*; FruB is variously called MTP (multiphosphoryl transfer protein), FPr, or DTP.

3. Expected step model (generic → KT2440)

#	Generic module step	Activity (GO/EC)	KT2440 gene	Status
1a	PEP-dependent fructose PTS phosphorelay (EI/HPr/EIIA)	PEP-protein phosphotransferase, EC 2.7.3.9; GO:0009401	fruB / PP_0793 / Q88PQ5	covered (but missing from metadata)
1b	Fructose-specific EIIB-EIIC transport + phosphorylation → F1P	fructose PTS EIIBC transporter, EC 2.7.1.202; GO:0022877	fruA / PP_0795 / Q88PQ3	covered
2	1-phosphofructokinase: F1P → FBP	1-phosphofructokinase, EC 2.7.1.56; GO:0008662	fruK / PP_0794 / Q88PQ4	covered

All three steps are **covered**. No step is a gap or "not expected." The known step relationship (PTS supplies F1P to FruK) holds and is reinforced by regulation: F1P is the sole physiological effector of the Cra/FruR repressor of the fruBKA operon (Chavarria et al. 2014,

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— regulatory detail that lies just outside the module boundary but confirms the operon logic.

4. Candidate genes and evidence

4.1 True module genes (high confidence)

- **fruA — PP_0795 (Q88PQ3)**. Fructose PTS EIIBC permease, EC 2.7.1.202, 580 aa (fused EIIB+EIIC). GO:0022877 (fructose PTS transporter activity), GO:0009401 (PTS). Direct KT2440 evidence: member of experimentally studied fruBKA operon

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Role: step 1b (import + phosphorylation to F1P). *Caveat:* candidate metadata bucket shown as ppu02060, correct; EC is specific, no over-mapping.

- **fruK — PP_0794 (Q88PQ4).** 1-phosphofructokinase, KEGG K00882, EC 2.7.1.56, 315 aa. GO:0008662 (1-phosphofructokinase activity) + ATP binding. **Role:** step 2 (F1P→FBP). *Caveat:* candidate metadata labels it generically "Phosphofructokinase" — this is **not** 6-phosphofructokinase (EC 2.7.1.11); the name should be curated to "1-phosphofructokinase" to avoid conflation with the (absent) Pfk.
- **fruB — PP_0793 (Q88PQ5). MISSING FROM CANDIDATE LIST.** Multiphosphoryl transfer protein (EI-HPr-EIIA^{Fru}), EC 2.7.3.9 (also EC 2.7.1.202 via KEGG K23993), 950 aa (multidomain, consistent with a fused relay). **Role:** step 1a (PEP phosphorelay). First gene of the fruBKA operon; contiguous UniProt accession (Q88PQ5) with fruK/fruA. Omitted only because it is annotated to ppu02060, not the ppu00051 seed bucket.

4.2 Over-propagated candidates (outside module boundary)

These are correctly annotated to their own functions but do **not** belong to the fructose-PTS module:

- **GDP-mannose / mannose branch:** `algA` (PP_1277), `PP_1776`, `cpsG` (PP_1777), `algC` (PP_5288), `gmd` (PP_1799), `rmd` (PP_1800) — mannose-6-P isomerase / GDP-mannose pyrophosphorylase / phosphomannomutase / GDP-mannose dehydratase. Function in GDP-mannose and cell-envelope sugar-nucleotide metabolism.
- **Alginate biosynthesis:** `algL` (PP_1281), `algG` (PP_1283), `alg44` (PP_1286), `alg8` (PP_1287), `algD` (PP_1288) — alginate polymerization/modification (largely cryptic under standard conditions in KT2440).
- **Other sugar catabolism:** `fucD` (PP_2831, L-fuconate dehydratase), `PP_2037` (aldolase, ppu00040).
- **Downstream-adjacent (FBP fate), still outside boundary:** `fba` (PP_4960, FBP aldolase EC 4.1.2.13) cleaves the module product FBP; `fbp` (PP_5040, FBPase EC 3.1.3.11) recycles FBP→F6P (essential for fructose→ED flux given Pfk absence); `tpiA` (PP_4715, triose-P isomerase) acts on triose products. These provide biological context but should not be counted as module members.

5. Gaps, ambiguities, and likely over-annotations

- **Metadata gap (highest priority):** `fruB`/PP_0793 is missing. Not a biological gap — the step is covered. Root cause: bucket seeded from ppu00051 while fruB maps to ppu02060.

- **Naming ambiguity:** `fruK`/PP_0794 carries the generic label "Phosphofructokinase," which risks conflation with 6-phosphofructokinase (Pfk, EC 2.7.1.11). KT2440 **lacks** a functional 6-PFK; the only F1P kinase is FruK (EC 2.7.1.56). Curate the name/EC explicitly.
- **Broad EC/GO on non-module genes:** `algA`/PP_1776 carry dual EC (2.7.7.13 + 5.3.1.8) and `algC` dual PMM/PGM activity — legitimate but irrelevant to this module; they are pathway-map over-propagation, not per-gene errors.
- **No lineage-specific replacement needed:** unlike some bacteria that split EI/HPr/EIIA into separate genes, KT2440 uses a single fused FruB — this is captured correctly by the 950-aa multidomain protein.

6. Module and GO-curation recommendations

Module step	Recommended mark	Rationale
1a PEP phosphorelay (<code>fruB</code>)	covered	<code>fruB</code> /PP_0793 present with EC 2.7.3.9; add to gene set
1b EIIBC transport → F1P (<code>fruA</code>)	covered	<code>fruA</code> /PP_0795, EC 2.7.1.202, direct evidence
2 1-PFK: F1P→FBP (<code>fruK</code>)	covered	<code>fruK</code> /PP_0794, EC 2.7.1.56, GO:0008662

- **Module status: covered / satisfiable.** No `gap`, `candidate_uncertain`, or `not_expected_in_target_taxon` calls are warranted for the three steps.
- **module_needs_revision (metadata only):** add `fruB`/PP_0793/Q88PQ5 to the module gene set; the generic module boundary itself is **correct** for KT2440.
- **Bucket caveat for curators:** do not treat KEGG ppu00051 membership as module membership — 15/18 candidates are out-of-scope map neighbors. Prefer KEGG modules/orthology (K23993, K02770, K00882) or the `fruBKA` operon as the seed for this module.
- **GO requests:** none needed — existing terms suffice (GO:0009401 PTS; GO:0022877 fructose PTS transporter activity; GO:0008662 1-phosphofructokinase activity; EC 2.7.3.9 for FruB). Recommend ensuring FruB carries a PEP-protein phosphotransferase / PTS term in the target proteome.

7. Genes to promote to full fetch-gene review

1. **fruB** / PP_0793 / Q88PQ5 — *highest priority*; currently missing; confirm EI-HPr-EIIA^{Fru} domain architecture and PTS relay role.
2. **fruK** / PP_0794 / Q88PQ4 — confirm/curate name to 1-phosphofructokinase (EC 2.7.1.56); disambiguate from absent 6-PFK.
3. **fruA** / PP_0795 / Q88PQ3 — confirm EIIBC fructose permease; verify F1P (not F6P) product specificity.

(Context-only, not module members: fbp/PP_5040 and fba/PP_4960 warrant a note re: EDEMP downstream fate but need not be promoted for this module.)

8. Evidence and open questions

Direct KT2440 experimental support: - Complete fructose PTS (PTS^{Fru}) and glucose-independent-of-PTS transport: Chavarría et al. 2012, *mBio* (

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- fruBKA operon identity and F1P as sole Cra/FruR effector: Chavarría et al. 2014, *Environ Microbiol*

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- ED-based downstream sugar catabolism, glucose peripheral pathways: del Castillo et al. 2007, *Environ Microbiol*

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Inferred (homology/database): exact FruB domain boundaries and F1P product specificity of FruA are assigned from KEGG orthology (K23993/K02770) and UniProt EC/GO; these are strong (fused-domain length and operon context are consistent) but not individually enzymatically re-verified in KT2440.

Open questions / resolving experiments: - Direct biochemical confirmation that FruA delivers fructose as F1P (vs F6P) in KT2440 (in vitro PTS assay). - Whether any secondary fructose uptake route operates in a Δ fruBKA background (growth/transport assay) — literature implies none. - Confirm FruK is strictly a 1-phosphofructokinase and cannot substitute for the missing 6-PFK.

Key references

- Chavarría M, Kleijn RJ, Sauer U, Pflüger-Grau K, de Lorenzo V. Regulatory tasks of the PEP-PTS of *Pseudomonas putida* in central carbon metabolism. *mBio* 2012.

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- Chavarría M, Durante-Rodríguez G, Krell T, et al. Fructose 1-phosphate is the one and only physiological effector of the Cra (FruR) regulator of *Pseudomonas putida*. *Environ Microbiol* 2014.

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- del Castillo T, Ramos JL, Rodríguez-Herva JJ, Fuhrer T, Sauer U, Duque E. Convergent peripheral pathways catalyze initial glucose catabolism in *Pseudomonas putida*. *J Bacteriol* 2007.

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- KEGG entries ppu:PP_0793 (K23993), ppu:PP_0794 (K00882), ppu:PP_0795 (K02770); UniProt Q88PQ5, Q88PQ4, Q88PQ3 (queried 2026).